# COMPREHENSIVE REPORT ON THE HYPERTENSION SCREENING PROGRAM PROPOSAL

## Program Rationale and Key Statistics

The district faces a rising hypertension challenge, necessitating the evaluation of a community screening program.

District Population: **30 lakh.**

Adults with Hypertension: **24%.**

Unaware of their condition (Undiagnosed): **40%.**

Villages without Regular BP Screening: **60%.**

Stroke Cases: Increased by **18%** in the last 5 years

## Determinants of Hypertension (Barriers)

Hypertension is driven by several factors locally:

**Lifestyle:** High-salt diets (10-11 g/day), low fruit intake, low physical activity, and common tobacco use in rural Maharashtra are key triggers.

**Occupation:** Daily wage laborers and farmers face high physical stress, long work hours (10-12 hrs/day), and have low health awareness, leading to missed check-ups.

**Income:** Low average income (INR 9,000) limits access to nutritious food and healthcare, causing villagers to skip screenings due to fear of losing a day's wage.

**Access Barriers:** 60% of villages lack regular BP screening due to distance, transport costs, and a shortage of trained health workers.

**Awareness:** Only 40% of people know about hypertension risks, leading to misconceptions like "BP is only for the rich" and delayed care.

# Demand and Supply Assessment

### Demand

The population might value early detection for preventing strokes, reducing hospital costs, and maintaining productivity. However, demand for screening varies by group:

| Group | Demand Level | Reasons |
|---|---|---|
| Adults >40 yrs | High | More risk, more awareness |
| Diabetic/obese persons | High | Know they are high-risk |
| Daily wage workers | Low | Time loss, income loss |
| Young adults | Low | Think they are healthy |
| Remote villages | Nil | No screening site available |
| People distrust health workers | Nil | Do not participate |

### Supply (Current Status)

Current supply involves mobile vans, ASHA/ANM support, and fixed points (PHC/HWC). However, the key weaknesses are:

- 60% of villages without regular screening.
- Staff shortage and irregular services.

# Microeconomics Evaluation (Cost-Benefit Analysis)

| Component | Cost Calculation | Total Cost (Year 1) |
|---|---|---|
| Training (Village Workers) | 50 workers × ₹5,000 (Training + Kit) | ₹2.50 Lakh |
| Worker Incentives | ₹10 per screening × 2.7 Lakh screenings | ₹27.00 Lakh |
| BP Machines | 100 units × ₹2,000 (Digital Omron) | ₹2.00 Lakh |
| Screening Camps Logistics | Logistics for district-wide monthly camps | ₹12.00 Lakh |
| IEC / Awareness | Posters, leaflets, announcements | ₹5.00 Lakh |
| Total Annual Program Cost | (Sum of all components) | ₹48.50 Lakh |

| Indicator | Value |
|---|---|
| Annual program cost | ₹48.50 Lakh |
| Annual medical savings | ₹259.00 Lakh (2.59 Cr) |
| Net benefit | ₹210.50 Lakh |
| Benefit-Cost Ratio (BCR) | 5.34 |

# SWOT Analysis

| Category | Item |
|---|---|
| Strengths | Mobile vans available |
| Strengths | ASHA/ANM support |
| Strengths | PHC/HWC facilities present |
| Strengths | BP machines available |
| Weaknesses | 60% villages without screening |
| Weaknesses | Staff shortage |
| Weaknesses | Irregular services |
| Weaknesses | Equipment problems |

# Advocacy and Equity Arguments

The program is necessary and highly implementable based on the following arguments for government officials and NGOs:

**Equity Argument:** It bridges health inequities by ensuring rural, poor, and daily wage workers—who currently have limited access—receive timely care. It also reduces gender-based barriers.

**Economic Argument: I**t reduces expensive hospital admissions (like for stroke) and is a cost-effective NCD intervention.

**Financial Protection:** By catching the disease early, it prevents catastrophic health expenditure that forces poor households into debt.

**Productivity:** Early detection and treatment prevent work absenteeism and preserve the economic productivity of daily wage laborers

# Planning, Budget, and Monitoring Recommendations

## PLAN A – FULL-SCALE DISTRICT-WIDE SCREENING

**Coverage target: 80% of all adults (18–65 years) = 14.4 lakh people in Year 1**

**Objective: Rapid detection of all undiagnosed hypertension cases and strong linkage.**

### 1. Population & Coverage Strategy

- Total adults = 18,00,000
- Screening target = 14,40,000 (80%)
- Includes:
  - All villages
  - Workplaces (factories, daily-wage clusters)
  - Schools/colleges for staff
  - Urban slums

### 2. Workforce Deployment

| Cadre | Role | Strength Needed |
|---|---|---|
| ASHA | Household listing, mobilization, follow-up | 1 per 1000 population → ~3000 ASHAs |
| ANM / MPW | BP screening, camp support | ~400–500 |
| Medical Officers | Confirmatory diagnosis, treatment initiation | 1 per PHC (~80–100 MOs) |
| Data Entry Operator | Enter NCD Portal data | 1 per block (12–15 DEOs) |
| District NCD Cell | Planning, supervision, M&E | Existing staff + 1 consultant |

### 3. Service Delivery Approach

#### A. Community Screening

- Door-to-door visits
- Workplace screening camps
- Outreach in weekly markets, bus stands
- Integration with VHNDs (Village Health Nutrition Days)

#### B. Confirmatory Care

- Daily NCD clinic at each PHC
- If BP ≥140/90 → confirm BP + basic blood/urine tests
- Risk scoring (age, diabetes, smoking)

#### C. Treatment Initiation

- Start first-line medication immediately
- Provide 30–60 days drug supply
- Educate about salt reduction, exercise

#### D. Follow-Up

- ASHA home visits monthly
- PHC BP check every 3 months
- Digital record update in NCD Portal

#### E. Referral System

- Uncontrolled patients → CHC
- Complications → District Hospital → Tertiary

### 4. Logistics Plan

- 2000–2500 BP apparatus (for ASHAs + camp
- IEC material for 1500 villages + urban wards
- Drug buffer stock for 3 months
- Tablets for DEOs

### 5. Timeline

| Month | Activity |
|---|---|
| 1 | Training + procurement |
| 2–7 | Mass screening phase |
| 3–12 | Confirmatory tests + treatment linkage |
| Continuous | Follow-up + monitoring |

### 6. Expected Outputs

- Screened: 14.4 lakh
- New hypertensives identified: ~1.38 lakh
- Linked to care: ~97,000
- On treatment: ~68,000
- Controlled after 1 year: ~34,000

### 7. Pros & Cons

Pros

- Rapid reduction in undiagnosed HTN
- Strong population-level impact
- Useful for political visibility and district performance ranking

Cons

- High cost (~₹27–28 crore)
- Heavy burden on workforce
- High logistics demand
- Risk of poor-quality data if rushed

# PLAN B – PILOT + PHASED SCALE-UP

**Coverage target: 30% of adults in Year 1 = 5.4 lakh people**

**Objective: Test feasibility, capacity, costs, and system readiness before expanding to whole district.**

**1. Population & Priority Strategy**

Screen only high-burden, high-risk areas, such as:

- Villages with no regular BP screening
- Areas with high stroke admissions
- Daily wage workers, construction sites
- Tribal or remote pockets

Adult population for pilot = 5,40,000

**2. Workforce Deployment**

| Cadre | Role | Strength Required |
|---|---|---|
| ASHA | Mobilization, follow-up | Only in selected pilot areas (~900 ASHAs) |
| ANM/MPW | Conduct screening | 100–150 personnel |
| Medical Officers | Confirmatory care | 30–40 MOs |
| DEO | NCD Portal reporting | 1 per pilot block (5–7 blocks) |

*(Workforce load is much lower compared to Plan A.)*

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**3. Service Delivery Approach**

**A. Screening Activities**

- Fixed-day weekly screening camps
- Outreach at worksites
- Opportunistic screening at PHCs, CHCs, District Hospital

**B. Confirmatory Pathway**

- Dedicated NCD OPD day once a week/Fortnight
- Priority care for screen-positives to prevent drop-out

**C.Treatment Initiation & Follow-Up**

- Provide 30–60 day drug supply
- ASHA follow-up in smaller catchment area
- Easier supervision since pilot area is limited

**4. Logistics**

- Fewer BP apparatus needed (~700–900)
- Manageable drug stock
- Lower IEC cost
- Data entry manageable by existing DEOs with small increase

**5. Timeline**

| Month | Activity |
|---|---|
| 1 | Training + micro-planning |
| 2–4 | Pilot screening (5.4 lakh adults) |
| 5 | Mid-term review (coverage, linkage, cost) |
| 6–12 | Follow-up + Treatment monitoring |
| Month 12 | Decision: Expand to full district (→ Plan A) |

**6. Expected Outputs**

- Screened: 5.4 lakh
- New hypertensives detected: ~52,000
- Linked to care: ~36,000
- On treatment: ~25,000
- Controlled after 1 year: ~12,500

**7. Pros & Cons**

Pros

- Lower cost (₹8–10 crore)
- Realistic for workforce
- Stronger quality control
- Allows program refinement before scaling
- Easier monitoring & supervision

Cons

- Fewer people reached in Year 1
- Benefits (stroke reduction) happen slower than Plan A

# BUDGET RECOMMENDATIONS

## A. Estimated Costs – Year 1 (80% Coverage Scenario)

| Component | Calculation | Cost (₹) |
|---|---|---|
| Screening | 14.4 lakh × ₹50 | **7.20 crore** |
| Confirmatory testing | 3.45 lakh × ₹200 | **6.91 crore** |
| Treatment for 1 year | 67,738 × ₹2,000 | **13.55 crore** |
| TOTAL | – | **₹27.66 crore** |

➡ **Approx: ₹28 crore needed for full-year, full-scale screening.**

## B. Pilot Budget Option (Recommended)

Screen **30% adults** (5.4 lakh):

| Component | Estimated Cost |
|---|---|
| Screening (₹50) | ₹2.7 crore |
| Confirmation | ₹2.1 crore |
| Treatment | ₹4.0 crore |
| **TOTAL PILOT** | **₹8–10 crore** |

➡ Safer, more realistic for Year 1.

➡ Helps measure feasibility + budget accuracy before scaling.

## C. Budget Line Items

1. **Human Resources**
   - ASHA incentive per screening event
   - Data entry operator (contractual)
   - Training costs for ANM/ASHA/MO
2. **Equipment**
   - Digital BP machines
   - Batteries/cuffs replacements
3. **Drugs & Consumables**
   - 1-year treatment stock
   - Diagnostic supplies
4. **IEC & Awareness**
   - Posters, banners, wall paintings
5. **Monitoring & IT**
   - Tablets for NCD portal
   - Internet/data packs
6. **Transport & Logistics**
   - PHC-to-village travel
   - Referral transport support

# PLAN A vs PLAN B

| Feature | PLAN A (Full Scale) | PLAN B (Pilot + Scale-Up) |
|---|---|---|
| Adults targeted | 14.4 lakh | 5.4 lakh |
| Coverage | 80% | 30% |
| Year-1 cost | ₹27–28 crore | ₹8–10 crore |
| Workforce load | Very high | Moderate |
| Data quality | Risky | Good |
| Linkage to care | Hard to manage | Easier |
| Best for | Rapid, district-wide impact | Smart, safe, sustainable rollout |
| Recommendation | Useful only if resources strong | **Best option for real-world districts** |

# KEY MONITORING INDICATORS

| Indicator | Formula | Target | Purpose |
|---|---|---|---|
| **1. Screening Coverage (%)** | Screened ÷ Eligible adults | ≥80% | Measures reach |
| **2. Screening Positivity (%)** | BP ≥140/90 ÷ Screened | ~24% | Detects WH accuracy & quality |
| **3. Confirmatory Linkage (%)** | Confirmed ÷ Screen-positive | ≥70% | Measures follow-up efficiency |
| **4. Treatment Initiation (%)** | Started meds ÷ Confirmed | ≥70% | Whether system start care |
| **5. Control Rate (%)** | Controlled cases ÷ On treatment | ≥50% | True outcom indicator |
| **6. Drug Stock-out Days** | Days PHC without BP meds | 0 days | System functioning |

## MONITORING TOOLS & TECHNIQUES

- STANDARD BP SCREENING REGISTER
- NCD DIGITAL PORTAL REPORTS
- LINE-LISTING OF SCREEN-POSITIVES
- REFERRAL TRACKING CARDS
- 
- CAMP MONITORING CHECKLIST
- MONTHLY HEAT MAP OF LOW-PERFORMING VILLAGES

## DATA SOURCES

- SCREENING REGISTERS
- NCD PORTAL ENTRIES
- PHC OPD REGISTERS
- ASHA MICRO-PLAN FOLLOW-UP LISTS
- DRUG STOCK REGISTERS

## MONITORING FREQUENCY

- BLOCK-LEVEL REVIEW → MONTHLY
- DISTRICT NCD REVIEW → QUARTERLY
- DASHBOARD SUBMITTED TO DHO → MONTHLY
- ANNUAL IMPACT REVIEW → YEARLY (COMPARE STROKES, ADMISSIONS)

## OUR RECOMMENDATION

**Plan B (pilot + phased scale-up) is recommended because it is financially feasible, easier to supervise, maintains higher quality, reduces risk of drug stock-outs, and allows refinement before district-wide implementation. Plan A should be adopted only after Plan B demonstrates ≥70% coverage, ≥60% linkage and good treatment continuity.**